

# Hansel Zhang

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## EDUCATION

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### Georgia Institute of Technology

Atlanta, GA

*B.S. Aerospace Engineering, Aug 2024 – May 2027*

- Current GPA 4.00 / 4.00
- Faculty Honors, 6 semesters

## MANUSCRIPTS IN PREPARATION

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1. **H. Zhang**, J. Skifstad, and G. Chou. “Causally Attributed Sparse Autoencoder Features in Vision–Language–Action Models.” *Manuscript in preparation for ICLR.*

## RESEARCH EXPERIENCE

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### Deformable Terrain Simulation for Rover Excavation

May 2026 – Present

*Aerospace Systems Design Laboratory, Georgia Institute of Technology · Advisor: J. McNabb*

- Wrote a material point method (MPM) solver from scratch using NVIDIA Warp API for high-fidelity GPU-accelerated deformable soil simulation.
- Built a Python interface coupling the Warp soil solver with NVIDIA PhysX, enabling bidirectional contact-force transfer between deformable terrain and articulated robots within a unified simulation loop.
- Implemented reinforcement learning pipeline for training rovers in the coupled simulator for excavation tasks.

### Feature-Level Interpretability of a Vision–Language–Action Model

May 2026 – Present

*Trustworthy Robotics Lab, Georgia Institute of Technology · Advisor: G. Chou*

- Used TopK sparse autoencoders to find interpretable features in OpenVLA, OpenVLA-OFT, MiniVLA, WorldVLA, and  $\pi 0.5$  that causally drive actions, verified by closed-loop ablation and additive steering across LIBERO suites.
- Ranked the recovered features by causal contribution to the emitted action without the need for human labeling.
- Identified general features firing at grasp and at release, alongside specialists tied to a single trajectory phase and specific skills.
- Showed that a near-constant baseline contributes to the action the policy commands, independent of the interpretable features.

### Symbolic Discovery of Structural Dynamics Equations

Jan 2026 – Present

*Moore Dynamics and Analytics Laboratory, Georgia Institute of Technology · Advisor: K. Moore*

- Extended deep symbolic regression, a transformer proposing candidate equations under a reinforcement learning objective, to recover the coupled equations of motion of multi-degree-of-freedom structures directly from vibration response.
- Replaced the binary expression-tree representation with n-ary trees carrying implicit constants, reducing tokens per equation and extending the model’s reach to more complex expressions.
- Reformulated the training reward around a work-energy balance residual, making small dissipative terms observable to model selection.

- Recovered governing equations from experimental data on both single-degree-of-freedom and multi-degree-of-freedom systems, including a coupled linear-nonlinear oscillator rig.

### **Multi-Agent Task Allocation for Lunar Surface Construction**

Jan – May 2026

*Aerospace Systems Design Laboratory, Georgia Institute of Technology · Advisor: J. McNabb*

- Adapted the consensus-based bundle algorithm (CBBA) to a discrete, agent-based simulation of lunar base construction, allocating tasks across a rover fleet under task precedence constraints.
- Preserved CBBA's convergence guarantee under the added constraints by enforcing diminishing marginal gain across each bundle and using agent IDs as deterministic tiebreakers.
- Encoded task precedence using a directed acyclic graph, partitioned into tiers where CBBA runs to convergence within each tier before moving on, avoiding restrictive time-window based CBBA formulations.
- Validated successful task assignment with 4–10 agents and 10–30 tasks.

### **OTHER EXPERIENCE**

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#### **Controls Engineer — Georgia Tech Propulsive Landers**

Jan 2025 – Present

*Georgia Institute of Technology*

- Built landing control stack for a propulsively landed vehicle in Rust to run on a Teensy 4.1.
- Implemented model predictive control for 6-DoF powered descent, re-linearizing nonlinear vehicle dynamics at each control step and constructing the quadratic program to be solved with OSQP under thrust-magnitude and gimbal-angle constraints.
- Wrote the onboard sensor processing firmware alongside the controller.
- Validated control stack against a simulator written by another team member, which used altered dynamics and wind disturbances absent from the controller's internal model.

#### **Undergraduate Teaching Assistant — CS 1371, Computing for Engineers**

Jan 2025 – Present

*College of Computing, Georgia Institute of Technology*

- Led weekly recitation for a section of ~20 students in a MATLAB-based introductory programming course for engineering majors.
- Prepared recitation material each week.
- Held weekly office hours and graded exams across five terms, including summer.

### **TECHNICAL SKILLS**

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**Languages:** Python, C++, Rust, MATLAB

**Frameworks and tools:** PyTorch, NVIDIA Warp, Isaac Sim

### **RELEVANT COURSEWORK**

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Robotics and Autonomy (Optimal Control, Model Predictive Control, Machine Learning, Reinforcement Learning, State Estimation *in progress*) · Control System Analysis and Design · System Dynamics and Vibration · Spacecraft Flight Dynamics · Probability and Statistics · Deformable Bodies · Structural Analysis · Differential Equations · Linear Algebra · Multivariable Calculus